House Price Prediction Using Machine Learning

Project Objective: The objective of this project is to develop a machine learning model that can predict house prices based on various property features such as area, number of bedrooms, bathrooms, stories, parking facilities, and other amenities. Accurate price prediction can help buyers, sellers, and real estate businesses make informed decisions.

Dataset Description

The dataset contains 545 house records with 13 features, including both numerical and categorical attributes. Important features include area, bedrooms, bathrooms, stories, parking, air conditioning, and furnishing status. The target variable is the house price.

Data Preprocessing

The dataset was explored for missing values and duplicates. No missing values were found. Categorical features such as main road, guestroom, basement, air-conditioning, and Pref area were converted into numerical form. Furnishing status was encoded using one-hot encoding to make it suitable for machine learning algorithms.

Models Used

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was split into training and testing sets using an 80:20 ratio.

Results

Linear Regression achieved the best performance with:

- MAE: 970,043
- RMSE: 1,324,507
- R^2 Score: 0.65

Random Forest Regressor achieved:

- MAE: 1,022,560
- RMSE: 1,401,497
- R^2 Score: 0.61

Therefore, Linear Regression performed slightly better on this dataset.

Key Insights

The analysis showed that area, number of bathrooms, air conditioning, stories, and parking spaces have the strongest influence on house prices. Most houses were concentrated in the mid-price range, while only a few properties had exceptionally high prices. The model was able to capture the overall pricing trend reasonably well.

Conclusion

This project successfully demonstrated the process of data cleaning, feature engineering, visualization, model training, and evaluation. The results indicate that machine learning can be effectively used for house price prediction, providing useful insights for real estate businesses and potential buyers.